Gaita: A RAG System for Personalized Computer Science Education

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GAITA: A RAG SYSTEM FOR PERSONALIZED COMPUTER SCIENCE EDUCATION

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Abstract

Navigating the changing landscape of Computer Science is more difficult than ever, for both newcomers to the field and seasoned professionals aiming to stay informed and up-to-date on trends.

With a plethora of online learning resources, self-learners are often left to haphazardly browse the web and enroll in the first or most affordable program they find—a consequence of choice overload that not only impedes their success and overall satisfaction, but also neglects to consider their unique perspectives and potential contributions to the field. It's extremely difficult to figure out which course(s) are right for you, and even harder to keep your goals in mind when worrying about prerequisites you might need. This can lead to enrolling in overly general introductory courses and ending up with a generic background in the field, ultimately making it harder to stand out in a competitive job market or make impactful contributions in your field.

We aim to solve this problem by developing Gaita, a Retrieval Augmented Generation (RAG) ChatBot that generates personalized learning pathways from our database of over 1,200 open-access Computer Science courses from Coursera and MIT OpenCourseWare tailored to individuals' specific backgrounds, needs, and ambitions.

Keywords: Computer Science Education, Personalized Education, Recommender Systems, Retrieval Augmented Generation

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Chapter 1

Introduction

1.1 Background and Motivation

Navigating the changing landscape of Computer Science is more difficult than ever, for both newcomers to the field and seasoned professionals aiming to stay informed and up-to-date on trends. With the increasing value of technical skills, Computer Science (CS) is increasingly interfacing with and transforming numerous industries, from healthcare to finance, and even the humanities. This shift has led to an increased demand for CS knowledge, as individuals across various fields recognize the value of acquiring technical expertise. While some individuals may explore university degree programs to understand common coursework and prerequisites, gaining access to these classes can be challenging due to high costs, rigid schedules, or other limitations.

Open-access courseware (OCW) has emerged as a valuable alternative, offering high-quality educational content for free or at a significantly reduced cost. These resources make it possible for self-learners to access university-level courses and specialized content from leading institutions, democratizing education and resources for those with limited academic opportunities. Platforms like Coursera, edX, and MIT OpenCourseWare have made it easier than ever to learn technical skills from anywhere in the world.

However, self-learners that turn to open-access courseware face an overwhelming number of courses and learning platforms. These OCW platforms provide a wealth of content, but with it comes an unintended side effect: choice overload. This abundance of options can make it difficult for learners to find courses that align with their backgrounds, goals, and existing skills. This can lead to their enrolling in overly general introductory courses and ending up with a generic background, ultimately dampening their unique interests, perspectives, and potential contributions to the field.

Moreover, the complexity of selecting the right learning pathway is enhanced by the challenge of understanding prerequisite requirements. Learners often lack clear guidance on which courses align with their current skill levels or goals, leading them to enroll in overly generic or irrelevant courses. This results in learners accumulating knowledge without a coherent learning pathway, ultimately hindering their learning success and overall satisfaction.

1.2 Problem Statement

The current landscape of online learning platforms leaves self-learners with two primary challenges: choosing the right courses among the overwhelming number of options and ensuring that their chosen courses align with their background and future goals. Users cannot find the course(s) suited to their needs using existing learning platforms. Existing course recommendation systems on open courseware platforms rely heavily on keyword-based search and popularity metrics, leading to generic recommendations that often prioritize widely-enrolled or popular courses. This lack of personalized guidance not only hinders learning success and satisfaction but also limits the effective use of their diverse backgrounds.

This gap highlights the need for a more advanced solution that not only recommends courses but also tailors learning pathways to each learner's specific needs, accounting for their prior knowledge, prerequisites, and goals.

1.3 Research Objectives

We aim to solve this problem by developing Gaita, a Retrieval Augmented Generation (RAG) chatbot that generates personalized learning pathways in Computer Science. Gaita leverages a database of over 1,200 open-access Computer Science courses from Coursera and MIT OpenCourseWare to provide tailored recommendations based on learners' specific backgrounds, needs, and ambitions.

As AI becomes increasingly integrated into every field, it is critical to equip individuals with the technical tools and knowledge to enhance their existing roles and passions, rather than focusing solely on technical professions. We aim to change how AI education can be harnessed to empower people in their diverse pursuits and provide personalized pathways that align with learners' unique goals and interests.

Chapter 2

Literature Review

2.1 Introduction

The growing demand for technical skills has led many learners to open-access courseware platforms like Coursera and edX. While these platforms offer a wealth of quality educational content, they also present challenges, such as navigating a vast selection of courses, identifying those that align with individual goals, and understanding prerequisite requirements.

This leads to choice overload, where too many options and insufficient guidance result in frustration and poor learning outcomes. Existing recommendation systems rely on vanilla algorithms that fail to capture the unique needs and diverse backgrounds of learners. Recent advancements in AI, particularly through Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), offer new opportunities to address these challenges. By providing more dynamic and context-aware recommendations, these models can create personalized experiences.

This chapter explores the obstacles faced by self-learners, the limitations of existing recommendation systems, and how LLMs can reshape personalized learning. It also identifies the key features of an effective system and the remaining gaps in current solutions.

2.2 AI is Changing Every Field

Advancements in Artificial Intelligence (AI) are transforming a wide range of industries [4] and driving the demand for technical skills and Computer Science knowledge [33]. From healthcare [36] to education [25] and creative industries [1], AI is enabling new innovations that require expertise in areas such as Data Science, Machine Learning, and Software Development. In sectors like healthcare, AI is being used to analyze large datasets, improve diagnostic accuracy [23], and enhance patient care [20]. Similarly, in finance, AI-powered algorithms are automating fraud detection and risk assessment [1]. These developments demonstrate the expanding influence of AI and highlight the increasing need for professionals with a strong technical foundation.

2.2.1 There is a Growing Demand for Technical Skills

As AI and technology continues to shape various fields, the demand for individuals with Computer Science knowledge has surged. According to recent studies, job postings requiring computer science and AI-related skills have increased significantly, emphasizing the rising value of these competencies in the job market [9] [38].

More professionals and learners are recognizing the value of technical expertise, both to remain competitive in their current roles and to explore new career paths [3]. Moreover, a growing number of students and early-career professionals are seeking knowledge in these areas [17]. This influx of learners from diverse backgrounds creates new challenges in identifying educational resources that are both relevant and appropriate for their various needs.

2.2.2 Traditional Education is not Enough

However, traditional educational programs are often not well-suited to meet this growing demand [28]. While university degrees offer comprehensive coursework, they are typically expensive and inflexible, making them inaccessible to many learners. These programs also struggle to meet the needs of continuing learners—those already in the workforce who wish to update their skills [24]. High tuition costs and rigid schedules limit access to these traditional pathways, leaving many learners seeking alternative ways to gain the necessary expertise [8].

2.2.3 MOOCs and their Limitations

For these reasons, an increasing number of learners are turning to more flexible and affordable options, such as open-access courseware and Massive Open Online Courses (MOOCs) [16] [29]. The rise of Massive Open Online Courses marked a significant shift in the accessibility of education. MOOCs were initially celebrated for their potential to democratize education, offering a plethora of high-quality courses from leading universities to anyone with internet connection.

In 2009, a meta-analysis by the U.S. Department of Education examined evidence-based practices in online learning and found that students who received some or all of their education online performed better on average than students receiving exclusively in-person instruction [27]. However, as the popularity of these platforms grew, studies revealed patterns of enrollment and attrition that highlighted several challenges. A preliminary case study on Australian open-access online education found that while MOOCs attracted a large and diverse group of enrollees, many learners struggled to stay engaged [15]. High attrition rates and low completion rates became characteristic of MOOCs, both emphasizing their capability for increasing access to education and raising concerns about their effectiveness in achieving long-term learning outcomes [7].

2.2.3.1 Low Retention Rates

High attrition and low completion rates in Massive Open Online Courses (MOOCs) are influenced by several factors, including learners enrolling in courses that may not align with their backgrounds or goals. This misalignment can stem from the overwhelming number of available courses, leading to choice overload and suboptimal decision-making. A study by Aldowah et al. identified key factors contributing to student dropout in MOOCs, emphasizing students' misalignment of prior experience, difficulty, and academic skills and abilities [13]. These elements highlight the importance of aligning course selection with individual learner profiles to enhance retention.

2.2.3.2 Choice Overload

The abundance of learning resources provided by MOOCs presents a double-edged sword: while it offers learners a wide variety of courses suited to different backgrounds and interests, it also overwhelms individuals with options that may not suit their specific needs or goals. Choice overload occurs when individuals are presented with so many options that the decision-making process becomes difficult, leading to potential dissatisfaction, indecision, or regret [6]. Research by Chernev et al. identifies choice set complexity, decision task difficulty, preference uncertainty, and the decision goals as key factors that contribute to this phenomenon. Moreover, the authors find that choice overload is directly correlated with satisfaction, confidence, regret, choice deferral, and switching likelihood. Moreover, choice overload results in deferral, frequent switching, and decreased satisfaction.

These effects are highly relevant to the issues of low completion and high attrition rates in MOOCs. When faced with overwhelming course options, learners may struggle to choose courses that align with their goals and abilities, often resulting in enrollment in courses that are not well-suited to their needs. This misalignment can increase the cognitive load on learners, making it harder to stay engaged and committed to a course over time. Additionally, choice overload may lead learners to defer or abandon their course selections altogether, contributing to high dropout rates. Frequent switching between courses, driven by dissatisfaction or regret, disrupts the learning process and can further hinder completion.

2.2.4 Recommender Systems and their Limitations

Recommendation systems emerged in the 1990s to combat choice overload [14]. They were initially developed as a response to the rapidly growing amount of information available on the internet. With an overwhelming number of media and online resources that the internet provided, users struggled to find relevant information among the vast options presented to them. Early recommendation systems were developed to help users navigate this abundance by filtering content based on user preferences, browsing behaviors, and prior interactions [26]. These systems leveraged collaborative filtering and content-based algorithms to make personalized suggestions, thus reducing cognitive load and enhancing user experience. By recommending items most relevant to users, these systems became crucial tools for managing information overload and have since evolved to support decision-making in a wide variety of domains, from movie [11] to news recommendations [34].

2.2.4.1 Recommender Systems for MOOCs

While recommender systems have become ubiquitous in many fields, their application in education, particularly in MOOCs, is a work in progress. Existing course recommendation systems lack true personalization, and rely instead on simple metrics like content-based or collaborative filtering [10] [37] [19]. This approach fails to capture the diversity seen in the learners, and do not account for the unique backgrounds, skills, and learning goals of individuals, leading to a misalignment between the courses recommended, and the intent and expertise of the learner [32].

2.2.4.2 Students Come from Varying Backgrounds

These systems undermine the diversity of learning experiences and perspectives that are essential for impactful contributions to both the learners' fields and to the broader field of Computer Science. Pushing widely-enrolled, trending courses to all learners, regardless of their goals or backgrounds, leads to users enrolling in overly general courses and ending up with a generic background. This, in turn, dampens individuals' diverse and unique interests and perspectives [2]. Moreover, the absence of personalized support and guidance not only limits the diversity of learning outcomes but also amplifies inequities, disadvantaging learners with non-traditional backgrounds and goals who lack the support needed to navigate their unique educational paths effectively [12].

The lack of personalization is especially problematic in the educational context, where relevance and appropriate alignment with the learner's background are crucial for successful educational outcomes [5]. Without tailored recommendations, learners struggle to find courses that are well-suited to their knowledge level, leading to frustration, disengagement, and ultimately, high attrition and low completion rates in MOOCs.

2.2.4.3 Bridging Students' Knowledge Gaps

A major challenge for learners using MOOCs, especially those from non-traditional backgrounds, is bridging their knowledge gap. Many learners enter these platforms with varying degrees of expertise, creating a need for personalized guidance on acquiring prerequisite knowledge. Without such guidance, learners often enroll in courses beyond their current skill levels and miss foundational knowledge crucial for success in advanced topics. Current recommendation systems on MOOC platforms frequently overlook this essential step, leading to frustration, high dropout rates, and unmet learning goals.

To address this challenge, some recent studies have explored recommendation systems that include prerequisite support to guide learners in their educational journeys. Parameswaran et al. (2009) highlight the importance of prerequisite recommendations, emphasizing that recommending courses based on a learner's prior knowledge can greatly enhance learning outcomes [30]. More recent studies propose implementing systems for assessing a learner's current knowledge level and recommending courses that build on students' prior knowledge to acquire more advanced skills [5]. Soto et al. (2020)

create a system to fill knowledge gaps in MOOCs by recommending parts of courses to students [39], and Wu et al. (2024) tackle this problem by proposing a Knowledge-Aware Meta-Concept that uses knowledge graphs for course recommendations [41]. Additionally, Gong et al. (2022) propose a novel approach called HinCRec-RL, which leverages reinforcement learning and network-based relationships between courses and knowledge concepts to provide more nuanced guidance to help learners fill specific knowledge gaps [32]. By aligning course recommendations with each learner's specific gaps and strengths, these systems can not only improve engagement and completion rates but also make it easier for learners to confidently advance to higher-level topics.

These challenges highlight the growing need for advanced recommendation systems capable of offering personalized pathways tailored to individual backgrounds and the evolving demands of various industries. For instance, learners from different fields—such as healthcare professionals aiming to learn data science or artists exploring computational creativity—require recommendations that recognize and address their specific contexts. Additionally, learners interested in data science applications may not need an in-depth knowledge of low-level programming languages, but instead will benefit more from foundational skills in data analysis, statistics, and introductory machine learning concepts. Without targeted guidance, learners may end up enrolling in irrelevant courses that do not align with their goals, which hinders both learner success and satisfaction when students struggle to progress meaningfully toward their objectives. In the context of MOOCs, this issue is further amplified: while large numbers of learners enroll in courses, only a small fraction complete them, often due to frustration with finding suitable courses and the absence of a structured pathway to support their learning journey

2.2.5 LLMs are Changing Recommender Systems

Current MOOC platforms often struggle to provide personalized learning experiences that align with each learner's unique goals and existing knowledge base. Traditional recommendation systems typically rely on popularity metrics or keyword matching, and do not effectively capture individual learner preferences. To address this issue, there has been a shift towards more sophisticated learning systems that can bridge the knowledge gap and adapt to each learner's background. Recent advancements in artificial intelligence, particularly through Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), offer new opportunities to address these challenges. By providing more dynamic and context-aware recommendations, these technologies have the potential to empower learners to pursue pathways that align with their interests and strengths and enhance the overall impact of online education [19].

LLMs, such as GPT-3, have demonstrated impressive general-purpose task-solving abilities, including the potential to approach recommendation tasks [22]. Unlike traditional search and recommender systems that rely on popularity metrics and keyword matching, LLMs can analyze natural language prompts and engage in conversational interactions to better understand learners' specific goals, prior knowledge, and preferred learning styles. This allows for a deeper, more nuanced understanding of user needs. For instance, Zhang et al. (2023) propose a deep learning-based course recommendation method for an online education platform that leverages word embeddings to better align recommendations with each individual's needs [42].

Retrieval Augmented Generation (RAG), originally developed for reducing hallucinations and enhancing language model performance on domain-specific tasks, show significant promise for recommender systems [35]. It integrates LLMs with external data sources: an approach that allows for the retrieval of contextually relevant information, which can then used to generate personalized recommendations. By combining the strengths of retrieval-based methods and generative models, RAG offers more up-to-date and dynamic recommendations, aligning educational content with each learner's unique goals and existing knowledge base. Rao et al. (2024) leverage RAG in their RAMO system to enhance MOOC recommendations, addressing the "cold start" problem by providing tailored course suggestions through a conversational interface [31]. However, with one-shot recommendation, it is unclear just how much of the user's experience can be captured.

2.3 What a System/Solution Should Include

To effectively address the challenges faced by Computer Science learners in MOOCs and other open-access education platforms, a course recommendation system must incorporate several key elements to provide a personalized and impactful learning experience.

2.3.1 Personalized Recommendations

A personalized learning strategy is crucial, especially for individuals entering a new field like Computer Science. Tailoring learning pathways to individual goals and aspirations enhances engagement and effectiveness. However, learners new to the field may lack sufficient information to independently chart an optimal learning path. Therefore, the system should offer personalized recommendations that consider each learner's background, prior knowledge, and unique goals, and create a tailored learning pathway that align with their specific needs. Prerequisite filling should be integrated to guide learners through appropriate foundational courses before advancing to more complex topics, reducing the likelihood of frustration and dropout. This will ensure that learners receive guidance suited to their current level and desired outcomes.

2.3.2 Goal-Oriented Learning Pathways

Laak and Aru (2024) emphasize that aligning educational technologies with learners' goals is crucial for effective personalized learning. They argue that modern educational objectives, such as developing general competencies and fostering learners' agency, should be central to personalized learning approaches, and that by starting with learners' goals, educational systems can better support learners' agency [21].

Chapter 3

Gaita Overview

3.1 Overview of Design and Purpose

3.1.1 Goals and Objectives

We introduce Gaita (Generative AI Teaching Assistant) with the primary goal of providing personalized learning pathways out of open access courses for Computer Science learners by leveraging advanced recommendation techniques. Gaita aims to address the challenge of choice overload by offering tailored recommendations that consider learners' background, prior knowledge, and personal goals. By providing more relevant course suggestions, Gaita seeks to enhance learner engagement, improve completion rates, and support users to succeed in their learning journey. Additionally, Gaita supports diverse learners, including those from other fields, by recommending Computer Science courses that are relevant to their specific area of study, ensuring the learning path aligns with their unique goals and background.

3.1.2 How Users Interact with Gaita to Generate Recommendations

Users interact with Gaita through a familiar chatbot interface, where they provide relevant information about their background and learning goals. Upon entering the platform, learners are prompted share details about their current knowledge, experience, and learning objectives (Figure 3.1).

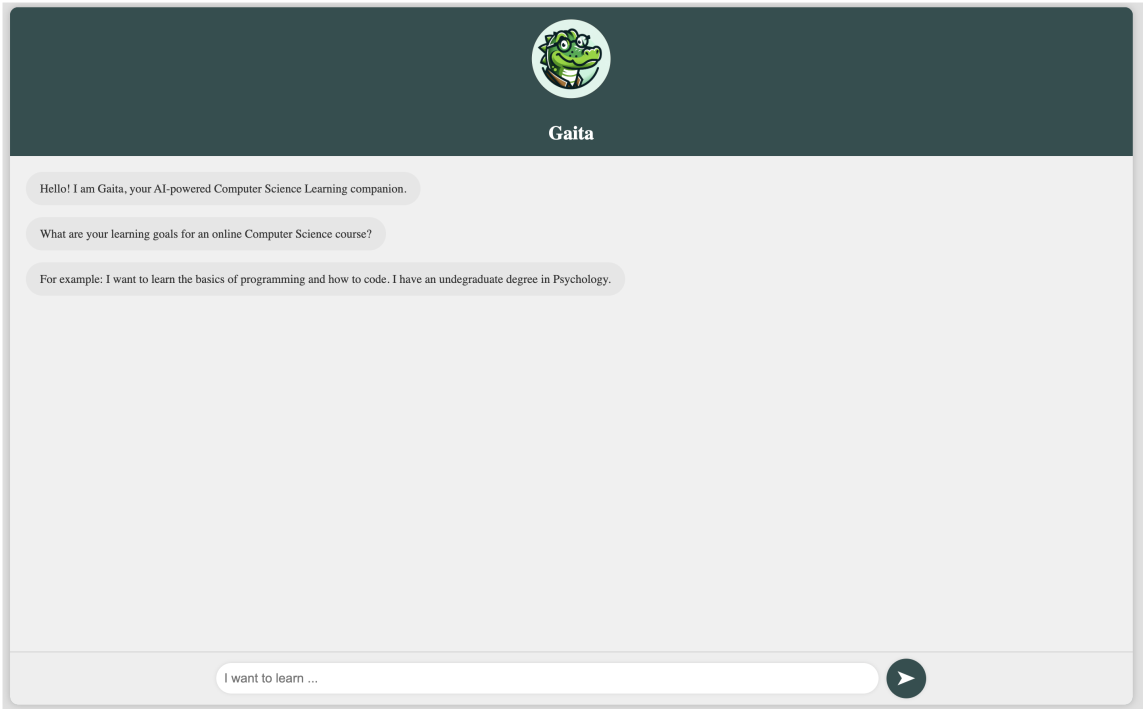

Figure 3.1: Gaita's Landing Page

Gaita responds with a list of relevant courses from its RAG database of open-access CS courses, along with descriptions of how they align with the user's prompt and relevant follow-up questions to assess the learner's experience with prerequisites for the recommended courses, which is where the iterative prompting comes into play. Based on the user's responses, Gaita suggests appropriate prerequisite courses to bridge any knowledge gaps (Figure 3.2.1).

The system iteratively refines its suggestions based on the learner's feedback, which ensures that the learning path remains relevant and aligned with the learner's evolving needs (Figure 3.2.2).

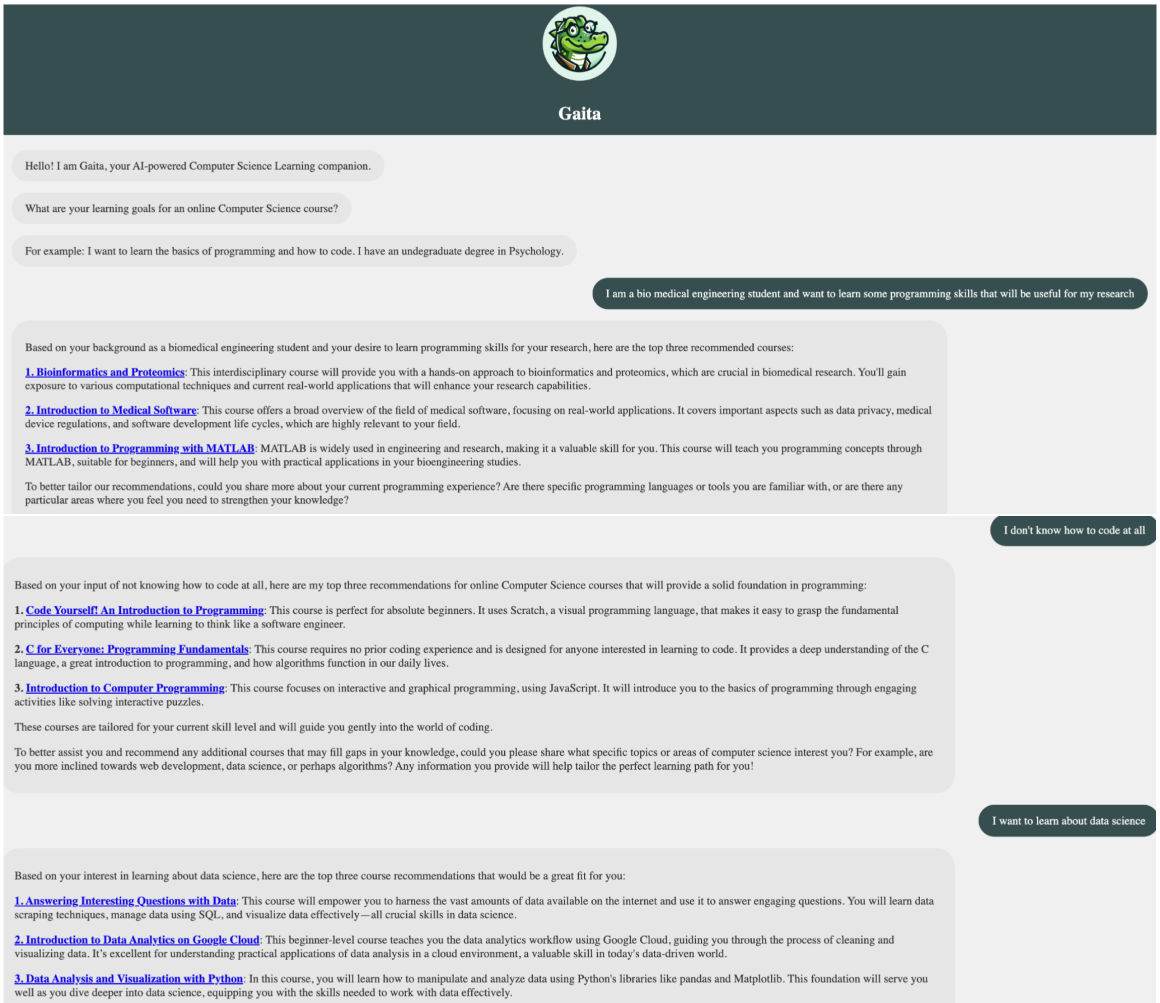

Figure 3.2: Sample Interaction

3.1.3 How Learners' Background and Goals Influence their Learning Path

The input provided by learners—such as their background, previous experience, and specific learning goals—directly shapes the learning paths generated by Gaita. For example, learners with a background in a different field or those transitioning into Computer Science will receive recommendations for foundational courses that address gaps in their knowledge. Meanwhile, learners with prior experience in certain areas will be guided toward more advanced or specialized content. By considering these inputs, Gaita ensures that the recommendations are not only relevant but also adaptable to each learner's prior knowledge.

3.2 Justification for Components

For the components chosen in Gaita, the decision to implement a chatbot interface was driven by the goal of enabling an interactive, iterative recommendation process that allows users to discover learning pathways tailored to their experience. The chatbot allows learners to input natural language prompts where they can detail their prior experience, share their goals, and build their course pathway throughout the conversation. This approach enables a more natural, adaptive experience compared to traditional forms of search.

Starting with learners' goals is a key component of the system's design. By focusing on their objectives from the outset, Gaita is able to generate more relevant and targeted course recommendations. Additionally, the system generates follow-up questions regarding user comfort with the prerequisites to help bridge any knowledge gaps with additional recommendations. Working backwards from the user's goals to bridge knowledge gaps ensures that the user does not take irrelevant introductory courses.

We chose to restrict the recommendations to courses from Coursera and MIT OpenCourseWare because both platforms are well-known and trusted, offering courses from reputable institutions. These platforms are widely recognized in the education space, and their courses are regularly updated and highly relevant, making them ideal sources for the initial course database. While centralizing CS resources is important, the primary focus of this project is on the recommendation pipeline and creating a personalized learning experience, rather than the data collection process itself. Our current database contains over 1,200 courses, and the modular nature of our system allows additional courses and platforms to be added in the future with minimal difficulty.

Although collaborative filtering using additional user data is a possibility, we decided not to scrape course reviews or LinkedIn data at this stage, as both are outside the scope of this project. While user reviews could provide valuable insights, they would introduce challenges related to data quality, filtering, and other ethical considerations. Similarly, scraping LinkedIn profiles raises privacy concerns and adds complexity, making it more appropriate for future iterations of the system if necessary.

3.3 Data Sources

Gaita uses open-access Computer Science courseware from Coursera and MIT OpenCourseWare, which provide high-quality courses from well-known institutions. These sources were chosen for their credibility and extensive offerings. The current database includes over 1,200 courses, collected via web crawling to efficiently aggregate up-to-date course data.

Chapter 4

System Design and Architecture

4.1 Architecture Diagram and Flow

Gaita's architecture diagram outlines the flow of data and how various components work together to generate personalized learning pathways for users. Here is a breakdown of the system:

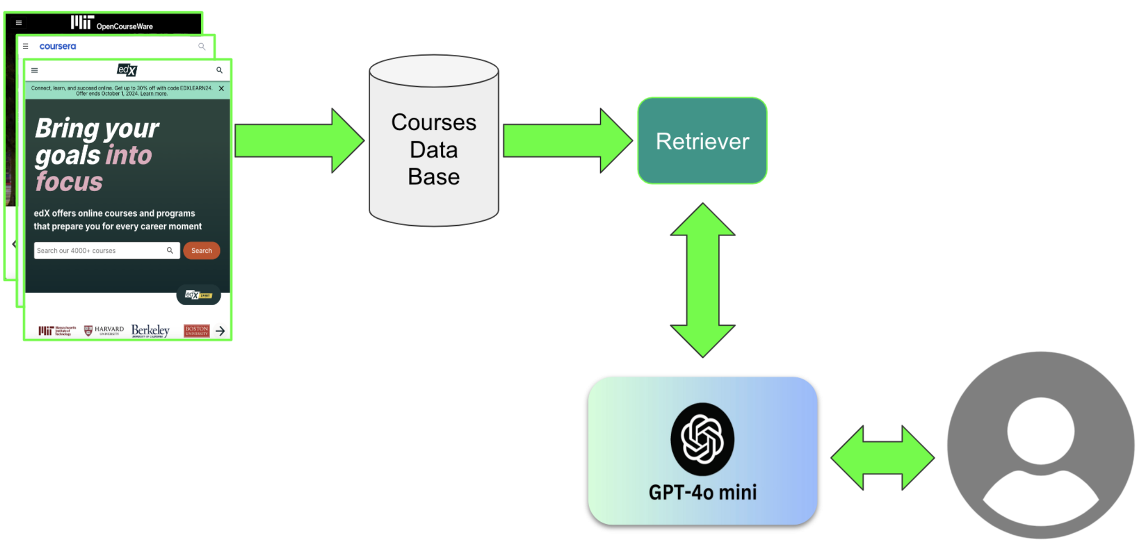

Figure 4.1: Gaita Architecture Diagram

1. **Scrape CS Courses from Coursera and MIT OCW:** Gaita collects course data from Coursera and MIT OpenCourseWare. These platforms provide high-quality Computer Science courses from leading institutions, ensuring a reliable and diverse range of learning materials.

2. **Courses Database:** The courses from these platforms are scraped and stored in a csv file. This database contains the titles, descriptions, and URLs of over 1,200 courses.

3. **Retriever:** The "Retriever" component is responsible for retrieving the most relevant courses based on the user's input. Using cosine similarity, the system compares the vectorized user prompt (which includes their goals and learning preferences) with the vectorized course data to identify the top 3 most relevant courses.

4. **GPT-4o Mini:** After retrieving the relevant courses, GPT-4o Mini serves the course recommendations along with descriptions of how each course fits the user's goals and follow-up questions to gauge the user's comfort with the prerequisites for the recommended courses.

5. **Iterative Prompting:** The system uses iterative prompting to create a personalized learning pathway. Based on the user's responses to the follow-up questions, Gaita assesses the user's current knowledge level and recommends prerequisite courses to fill any knowledge gaps.

6. **User Interaction:** The user interacts with Gaita through a simple and intuitive chatbot interface, where they can provide information about their background and learning goals. This interaction drives the entire recommendation pipeline, ensuring that the system continuously adapts to the learner's evolving needs.

7. **Deployed via Render:** Gaita is deployed on Render and is accessible at

https://gaita.onrender.com [1].

4.2 Components and Technical Tools

The following components and technical tools were used in the development of Gaita:

4.2.1 Web Crawling and Database Creation

Web crawling is employed to collect course data from Coursera and MIT OpenCourseWare, which are both well-established platforms offering high-quality educational resources. Web crawling ensures the system has access to up-to-date content, providing a reliable external data source for generating recommendations. This data is used to create a comprehensive database of over 1,200 Computer Science courses. Our database consists of the course title, description, and URL. OpenAI's text-embedding-3-small model was used to vectorize the course titles and descriptions in our database.

4.2.2 RAG-based Recommendation System

The recommendation system is based on a Retrieval-Augmented Generation (RAG) approach. After receiving the user input, Cosine similarity is applied to compare the vectorized user input (i.e., their goals) with the vectorized course database. The top 3 most relevant courses are retrieved, and GPT-4o Mini is employed to serve the recommendations with descriptions of how the courses meet the user's learning

[1] Important: If you encounter a 502 Bad Gateway Error, this is likely because the server is currently spinning up again. Render services can sometimes take a few minutes to become fully operational, so please wait a few minutes before refreshing the page.

objectives.

4.2.3 Turning Recommendations into Pathways

Gaita dynamically creates a learning path through iterative prompting. After outputting course recommendations, the system generates follow-up questions to assess the learner's prior knowledge and experience. Based on the responses, prerequisite courses are recommended to fill any knowledge gaps, ensuring the learner follows a coherent learning path that aligns with their goals. This iterative process helps to dynamically create a personalized learning pathway.

Chapter 5

Evaluation and Results

5.1 Qualitative Analysis of System Success

For the qualitative analysis of Gaita, user feedback was gathered through user interviews, which provided valuable insights into the system's effectiveness. The primary source of user data came from family and friends, who interacted with Gaita in a controlled, informal setting and provided insights on its usability and relevance of its recommendations. Additionally, I presented a demo of Gaita to a small group of professionals in the fields of education, machine learning, and human computer interaction, where they provided additional feedback on the system's performance and user interface. This provided initial impressions and qualitative feedback on the usability and relevance of Gaita, as well as important observations regarding user engagement, the clarity of the course recommendations, and the overall user experience.

5.1.1 Use of Open Courseware

One of the key strengths of Gaita is its exclusive use of open-access courses, which has been well-received by users. Users appreciated that the system recommends only freely available educational resources, as this helps bridge accessibility inequities in education. This ensures that high-quality learning materials are accessible to anyone, regardless of their financial situation or geographic location. The emphasis on open access content supports our mission of enabling users from diverse backgrounds to pursue their learning goals without the barrier of cost.

5.1.2 Personalization and Relevance of Recommendations

One of the primary goals of Gaita is to provide personalized learning pathways that align with users' goals, backgrounds, and knowledge levels. The system has shown significant success in this area. Based on user feedback, learners feel that the courses recommended by Gaita align well with their academic or career aspirations. Test users expressed satisfaction with the clarity of the course descriptions and how each recommendation was linked to their specific goals. They found the system's iterative prompting feature, which asks follow-up questions to determine prerequisite knowledge, to be intuitive and helpful in bridging knowledge gaps.

5.1.3 User Experience and Interface

The user experience and interface of Gaita were designed to be intuitive and simple, leveraging users' familiarity with existing messaging platforms. Users reported that the ability to input prompts in natural language made the system easier to use compared to traditional search systems found on other platforms. They found the conversational approach not only simpler but also more effective, as it captured more context and provided more nuanced, personalized recommendations than existing search-based systems on open courseware platforms.

Additionally, users found the follow-up questions regarding prerequisites are both relevant and helpful. These questions were intentionally designed to be easy to respond to, and enable the system to refine its recommendations and guide learners effectively through the course selection process. Additional feedback indicated that users appreciated the simplicity and clarity of these questions and found the experience more engaging.

5.1.4 System Performance and Reliability

Gaita has demonstrated strong performance in terms of both speed and reliability. The integration of RAG (Retrieval-Augmented Generation), which involves vectorizing courses and comparing them to the user's input using cosine similarity, has proven efficient and effective in generating relevant course recommendations. A key advantage of using RAG is its ability to mitigate the issue of hallucinations, a common challenge with Large Language Models (LLMs). Unlike vanilla LLMs, which sometimes generate non-existent or irrelevant course recommendations, Gaita ensures that only valid courses are suggested. The GPT-4o Mini component has also met expectations, consistently generating coherent and relevant course descriptions. These descriptions effectively explain how the recommended courses align with the user's background and goals, providing clear context for each suggestion. Overall, Gaita's performance has been reliable and accurate.

5.2 Qualitative Analysis of System Weaknesses

5.2.1 Inconsistent Formatting of Recommendations

One weakness identified in Gaita is the inconsistent formatting of the recommendations. At times, course titles or questions are boldfaced, while other times they are not, which leads to a lack of visual consistency. Additionally, the number of newlines between paragraphs can vary, due to the non-deterministic nature of the LLMs used. Users have provided feedback that they find the inconsistent formatting distracting. They expressed a preference for keeping the questions boldfaced and making the formatting more uniform throughout the system to enhance readability and user experience.

5.2.2 Difficulty Handling User Knowledge Claims

Gaita sometimes struggles when users explicitly state their prior knowledge, such as "I am familiar with coding in Python." In these instances, Gaita may still recommend introductory Python courses, even though the user already has familiarity with the topic. Users have noted that Gaita works best when they specify what they don't know or what they want to learn, rather than stating what they already know. Enhancing the system's ability to interpret knowledge assertions and using them as a stop condition would prevent unnecessary recommendations and lead to more effective pathway recommendations.

5.2.3 Dependency on User-Provided Data

Gaita depends on the data users provide about their backgrounds and learning goals to generate personalized recommendations. This introduces the risk of incomplete or inaccurate information. If users do not offer sufficient context or lack clarity about their needs, the recommendations may not be as effective. To improve the accuracy of the system, a more comprehensive data collection approach, such as connecting to user profiles or having users fill out an assessment form (with consent), could enhance the quality and relevance of the recommendations.

5.2.4 Oversimplification of Prerequisite Mapping

The iterative prompting system helps users identify prerequisite courses, but it carries the risk of oversimplifying the complexity of learning paths. For example, some advanced courses may require a deep understanding of a broad set of topics, not just a few prerequisites.

5.2.5 Limited Ability to Suggest Learning Paths beyond Course Recommendations

While Gaita provides excellent course recommendations, it currently does not offer alternative learning paths that could incorporate resources beyond courses, such as textbooks, projects, or practice exercises. A more comprehensive approach to personalized learning could involve suggesting supplementary materials to enhance the learning experience and better support skill development.

Chapter 6

Discussion

6.1 Absence of Quantitative Evaluation

In evaluating Gaita, we opted not to conduct a quantitative evaluation due to the absence of established baselines or existing systems for evaluating RAG (Retrieval-Augmented Generation) chatbots. Additionally, it would not be logical to compare Gaita against traditional search and recommendation engines, such as course search functionalities on platforms like Coursera, because Gaita operates with a fundamentally different input-output structure. While platforms like Coursera rely on traditional search and popularity-based recommendations, Gaita processes natural language input and tailors recommendations through a conversational interface, which makes a direct comparison inappropriate. [1].

A second evaluation we considered was to ask users to compare Gaita against other LLM chat systems, such as Gemini and GPT-4 Mini, using Likert scale surveys and metrics like precision at k=3. This evaluation would have included a 3x3 Latin square design to randomize the order of results and ensure balanced comparisons across systems. However, this approach was not carried out because it was not a suitable comparison. While Gaita shares a conversational interface with other LLMs, its

[1] In a prior project, I compared vanilla course search results on Coursera with an search interface that accepted natural language input on the same dataset, and found that the latter offered more context-aware and relevant recommendations [40].

iterative prompting feature, which tailors follow-up questions to assess prerequisite knowledge, is difficult to compare. Evaluating each system's effectiveness would require considering the entire conversation chain, which is challenging due to the complex, dynamic nature of conversational interactions. This iterative nature of Gaita's prompting is a key component that ensures that the system continuously adapts to the user's evolving needs, something that static systems or even other LLMs may struggle to achieve. Furthermore, traditional evaluation metrics, such as precision at 3, are designed for static outputs and do not account for the iterative and context-dependent aspects of conversation. This complexity makes it difficult to apply standard evaluation methods to conversational recommender systems.

It is important to note that conversational recommender systems (CRS) require an evaluation methodology that accounts for the iterative, context-dependent nature of user interactions, which is often overlooked by traditional evaluation metrics [18]. Unlike conventional recommenders that provide static lists of recommendations, CRSs interact with users through dynamic dialogues, making it essential to assess not only the accuracy of recommendations but also the overall user satisfaction and engagement. Jannach emphasizes that, given the interactive nature of CRSs, traditional metrics focused on algorithmic accuracy are insufficient on their own. Instead, the author advocates for a mixed-methods approach that combines objective, computational assessments with subjective, perception-oriented evaluations. Jannach argues that effective CRS evaluations must include measures of user satisfaction, engagement, and the quality of the interaction, all of which are integral to understanding how well the system meets user needs and provides value.

Comparing Gaita with standard recommendation systems or LLMs would not account for its unique features, such as tailored follow-up questions to assess prerequisite knowledge. Therefore, a qualitative evaluation focused on user feedback and system usability was more practical and effective for understanding Gaita's strengths and areas for improvement. While a quantitative approach was considered, qualitative insights offer deeper understanding of Gaita's user experience, the relevance of its recommendations, and opportunities for refinement in future iterations to better serve learners.

6.2 Iterative Prompting

To enhance Gaita's effectiveness, improving the iterative prompting feature is a priority. We plan to introduce a clear stop condition where Gaita stops recommending prerequisite courses once the user is adequately prepared. Additionally, we aim to improve the system's ability to respond more accurately to user knowledge claims, potentially by adding it to a user profile. By improving the system's understanding of user input, Gaita can offer more than just relevant course recommendations and become a better learning assisstant. Furthermore, the nuance of prerequisite recommendations can be enhanced, ensuring that users are guided through the necessary foundational steps, particularly for more advanced learning paths.

6.3 User Profiles and Progress Tracking

In order to offer a more personalized and adaptive learning experience, we plan to implement functionality for users to create and maintain a profile, allowing Gaita to store and remember key information about their learning history, preferences, and goals. This profile could be enhanced by incorporating data from surveys or external sources like LinkedIn profiles (with consent), providing more context on the user's professional background and specific learning needs. By integrating such data, Gaita can offer even more tailored recommendations based on the user's expertise and career aspirations. Introducing user profiles would enable the system to track a user's progress over time and adjust learning paths accordingly. For example, if a user finds a particular course too challenging or not engaging enough, Gaita could modify future course recommendations to better align with their current level of understanding and interests.

6.4 Integrating with Existing Platforms

A key component of expanding Gaita's impact is integrating it with existing open courseware platforms, such as Coursera and MIT OpenCourseWare. Gaita's modular design allows for easy integration by swapping out the database to pull course offerings from specific platforms or individual universities. Collaborating with universities to incorporate their course catalogs into Gaita would enable the system to provide tailored learning pathways for students, helping bridge gaps in traditional education systems and supporting learners from non-traditional backgrounds. AI-driven personalized course planning could help university students with educational planning, particularly when advisors may not provide in-depth guidance on their specific goals. Students are more likely to be open about their objectives with a chatbot, allowing for deeper, ongoing conversations, including tracking progress and generating alternative plans.

Additionally, Gaita could offer backup course suggestions if a student is unable to enroll in a desired class, making the learning process more flexible and responsive to individual needs.

6.5 Expanding Learning Resources

To improve Gaita's recommendations, expanding the database to include more courses is important. Additionally, broadening the scope of learning resources beyond courses—such as incorporating readings, hands-on projects, and practice exercises—would offer a more holistic and personalized learning experience. By providing a variety of learning methods, Gaita can cater to different learning styles and needs, supporting learners in a more comprehensive manner and enhancing the flexibility of educational pathways it offers.

6.6 Future Evaluation

As we continue to develop Gaita, future evaluations will focus on obtaining a more diverse sample of user data. This will help to ensure that Gaita's effectiveness is not limited by a narrow user base, allowing for insights into how it performs across different demographics, learning goals, and educational backgrounds. Additionally, more structured user interviews and feedback will be conducted once Gaita reaches a more refined state. These interviews will help gather deeper insights into the usability of the system and how its recommendations meet users' needs. As the system gains more traction, quantitative metrics, such as user engagement, enrollment, and completion rates will become relevant and can be incorporated into future evaluations. While these aspects are currently out of scope, as the focus was on establishing the system's foundation, they will be important to consider in future phases of development.

6.7 Ethical Considerations

Designing personalized learning pathways introduces several ethical considerations, particularly around user privacy. While Gaita currently does not collect any user data, once user profiles are integrated, it will be essential to ensure that sensitive information, such as personal learning history and career goals, is securely managed and protected. The handling of user data must comply with privacy regulations and respect user consent, ensuring transparency about what data is collected and how it is used. As Gaita evolves, careful attention will be required to balance the benefits of personalization with the ethical responsibility to safeguard user information privacy.

Chapter 7
Conclusion

7.1 Key Contributions

This work introduces Gaita, a novel personalized learning system that leverages conversational AI and Retrieval-Augmented Generation (RAG) to deliver dynamic, context-aware course recommendations. Unlike traditional search algorithms that return results based solely on relevance or other RAG models that aim to prevent hallucination, Gaita harnesses the interpretive power of Large Language Models (LLMs) to offer personalized learning paths that actively engage with the learner's unique background and aspirations. A key contribution is Gaita's iterative prompting mechanism, which tailors follow-up questions to assess prerequisite knowledge. While iterative prompting has been explored in various contexts, Gaita is, to our knowledge, the first system to implement it specifically for bridging knowledge gaps and recommending prerequisite courses within personalized learning pathways. This dynamic approach addresses a critical gap in traditional recommendation systems, which often fail to consider the evolving needs of users.

Moreover, Gaita's unique ability to take in natural language input is among the first implementations for open courseware, allowing for more nuanced user understanding and an improved user experience. This conversational approach allows Gaita to better interpret user needs and aspirations, offering more personalized and relevant course recommendations than traditional search systems.

Furthermore, Gaita's modular design allows for easy integration with existing open courseware platforms, such as Coursera and MIT OpenCourseWare, as well as individual university course catalogs. This flexibility expands the range of resources available for personalized recommendations, positioning Gaita as a scalable solution that can cater to learners, ranging from university students to lifelong learners, by offering tailored educational paths across different platforms. Ultimately, this work lays the groundwork for further advancements in personalized learning systems, by incorporating conversational AI and RAG to offer an innovative approach to helping learners navigate complex educational landscapes.

7.2 Impact and Implications

The development of Gaita represents a significant step forward in the application of Artificial Intelligence (AI) and Information Retrieval (IR) technologies in education. By leveraging conversational AI, Gaita transforms how learners interact with the vast array of available educational resources, by tailoring their learning journeys to their unique backgrounds and goals. This approach not only enhances the learning experience for individual users but also has broader implications for the future of EdTech and personalized learning. Gaita's use of natural language processing enables conversational interactions, which helps bridge the gap between traditional search systems and the personalized, adaptive systems learners increasingly expect.

In the broader context of EdTech, Gaita's integration with open courseware platforms supports the democratization of education by helping learners from diverse backgrounds, including nontraditional and lifelong learners, and those underserved by conventional systems. By creating personalized learning pathways, Gaita engages users and ensures their learning experience is continuously adapted and refined. This approach showcases the potential of AI-driven systems to not only recommend relevant content but also interact with learners, supporting their pursuit of knowledge and enhancing the overall educational experience.

Ultimately, Gaita demonstrates the potential for Artificial Intelligence and Information Retrieval technologies for creating more accessible, inclusive, and effective education systems. As these technologies continue to evolve, they have the potential to transform the future of education by personalizing learning experiences at scale, empowering learners from diverse backgrounds, and ensuring that education is tailored to the unique needs, backgrounds, and aspirations of every individual.

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Appendix A

More Sample Interactions

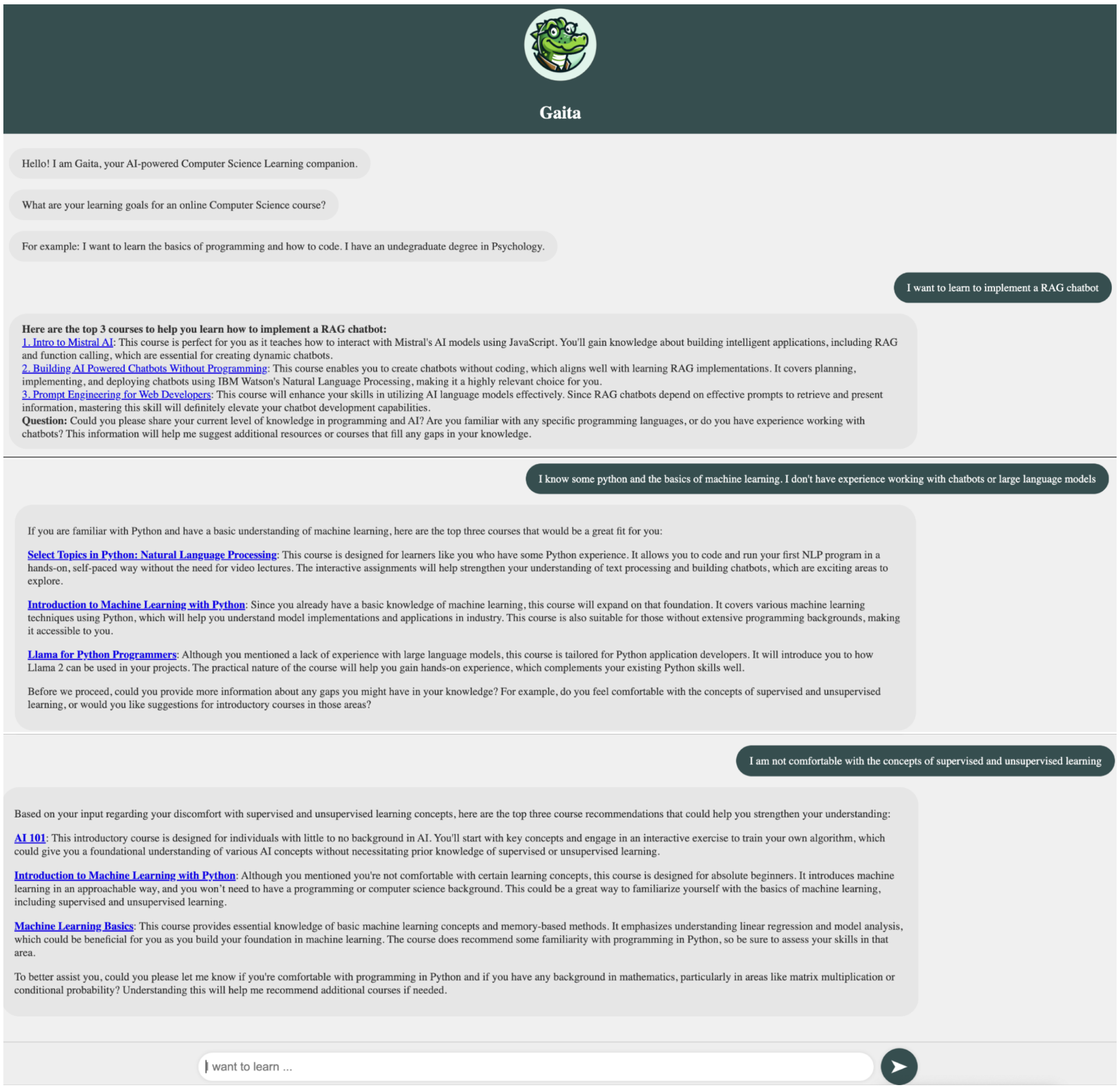

Figure A.1: Sample Interaction